

K-FIELD: A HIGH SCHOOL DESCRIPTION

Cell3 geometry, the Lucas Group, seven-sheet organization, graph structure, and cross-domain dimensional collapse

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Purpose of this reference

This paper explains the K-Field for high school science students. It is designed to stand alone. The geometry, tables, node addresses, and figures are generated from one authoritative Cell3 basis. The figures use equal physical scaling so that one nanometre is represented consistently along every displayed coordinate axis.

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1. The main idea of the K-Field

The K-Field describes space as an organized geometric environment rather than a completely featureless background. A useful analogy is graph paper: a drawing may look continuous, but the hidden grid supplies directions, spacing, and reference positions. The K-Field extends that idea into a three-dimensional, skewed, recursively repeating structure.

The framework begins with one measured local shape called Cell3. Cell3 is expanded into the Lucas Group, divided into exact nodes and sheets, and connected as a graph. Physical quantities are then read as different responses to the same underlying geometry. The central educational idea is simple:

one underlying geometry - many physical appearances

The sections that follow show how that idea is built mathematically from actual Cell3 data.

2. Cell3: the base geometry

Cell3 is a parallelepiped: a slanted three-dimensional box formed from three directed edge vectors. The three vectors have different lengths and do not meet at right angles. They are fixed in the International Celestial Reference System, or ICRS, which is an astronomical coordinate frame.

Vector	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	Length (nm)
a	0.327501242070	-0.054897007886	-0.037487502896	0.334179679073
b	0.035330775844	0.207319850441	-0.547401085392	0.586410890415
c	-0.010129317627	-0.389787702364	-0.689734408363	0.792319765038

The included angles are $\text{angle}(a,b) = 83.933483012$ deg, $\text{angle}(a,c) = 80.448130018$ deg, and $\text{angle}(b,c) = 50.363231814$ deg. The ordered determinant is $-0.117914891917 \text{ nm}^3$, so the ordered basis is left-handed. The physical volume is its absolute value, $0.117914891917 \text{ nm}^3$.

Authoritative Cell3 in ICRS coordinates Equal scale on x, y, and z axes

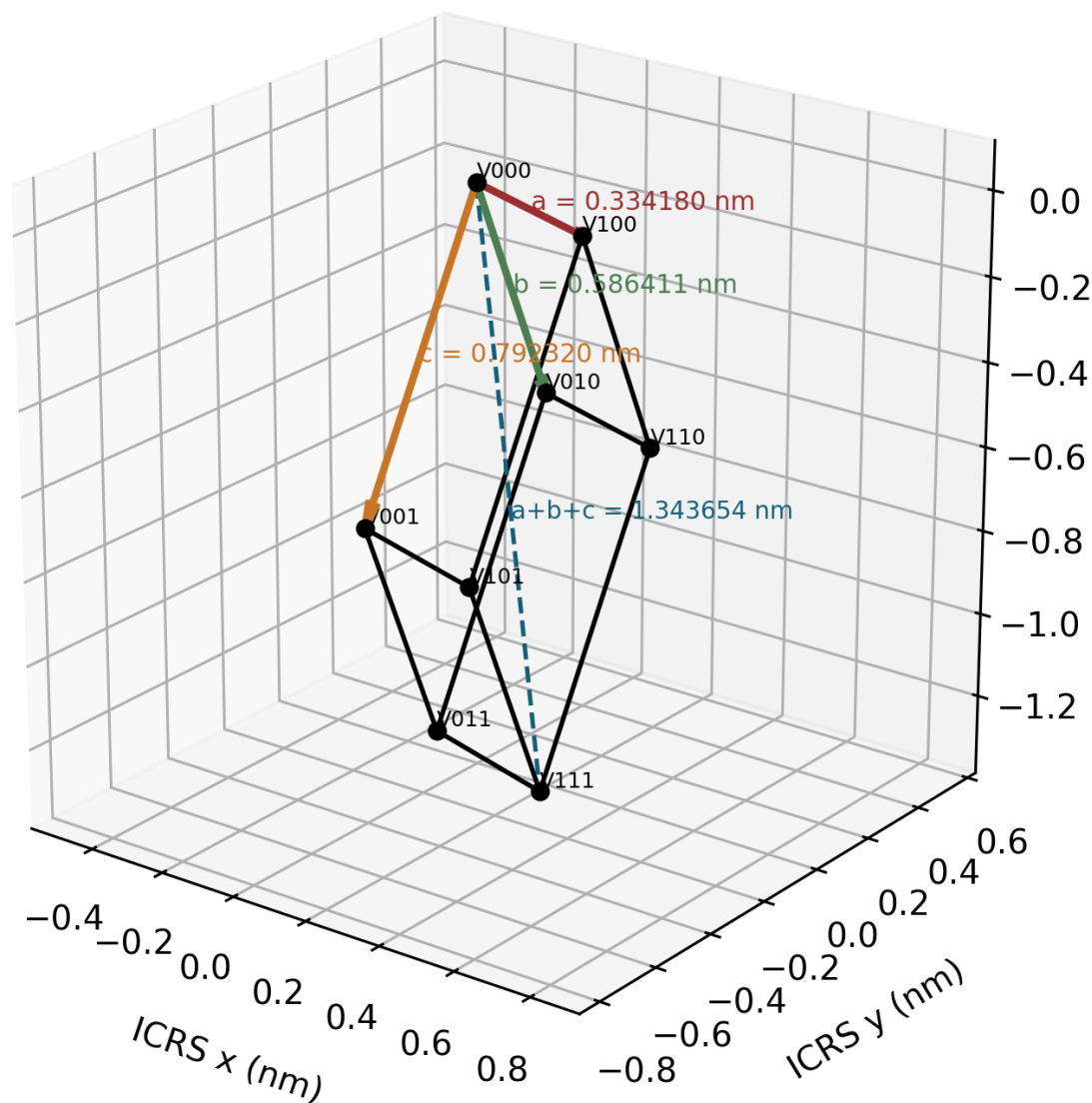


Figure 1. Authoritative Cell3 in ICRS Cartesian coordinates. All three axes use the same physical scale, so the geometry is not stretched to fill the page. Every vertex, edge, vector, and the body diagonal is generated from the numerical basis.

3. Exact face and plane systems

Cell3 has three pairs of opposite faces. Each face pair has one exact area and one exact perpendicular separation. The same information can be described through the reciprocal plane families. Direct edges describe displacement; reciprocal planes describe coordinate level sets and phase layers.

Family	Face area (nm ²)	Opposite-face spacing (nm)	Geometric meaning
ab	0.194869165470	0.605097741515	faces spanned by a and b
ac	0.261106235800	0.451597379724	faces spanned by a and c
bc	0.357809539129	0.329546529709	faces spanned by b and c

Three exact opposite Cell3 face-plane families All panels use the same nanometre scale

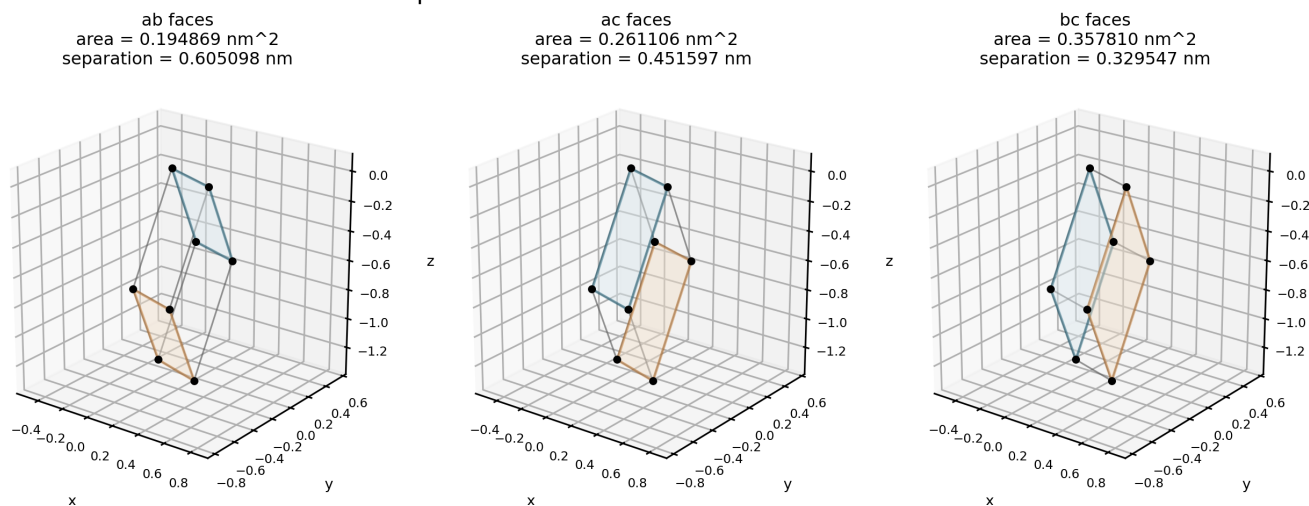


Figure 2. The three exact opposite face-plane families. Every panel uses identical nanometre scaling and the same camera direction. The unequal apparent sizes are physical consequences of the Cell3 geometry, not drawing distortion.

The face areas also appear in edge-aligned orthographic projections. For a view along edge i , the projected length of edge j is A_{ij} divided by the length of i . Therefore, the three projected side ratios reproduce the three face-area ratios exactly.

Three exact edge-aligned orthographic projections All panels use the same physical scale

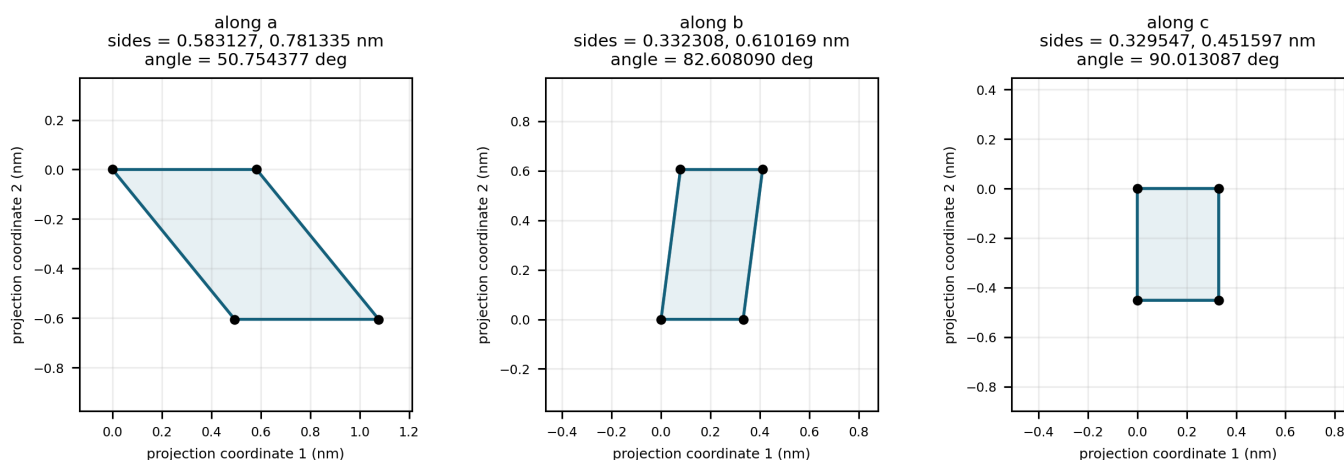


Figure 3. Exact edge-aligned orthographic Cell3 projections. All three panels use the same nanometre scale. The displayed side lengths and included angles are computed directly from the authoritative vectors.

4. The Lucas Group and recursive scales

The Lucas Group contains three Cell3 units along each primitive direction, giving $3 \times 3 \times 3 = 27$ complete Cell3 volumes. Its outer basis is $3B$. Selecting one Cell3 from the group is an exact division by 3. After that, a selected Cell3 is divided repeatedly by 2.

$3B \rightarrow B \rightarrow B/2 \rightarrow B/4 \rightarrow B/8 \rightarrow \dots$

The first divisor and later divisors have different jobs. Division by 3 selects a constituent Cell3 from the group. Division by 2 performs internal recursive refinement.

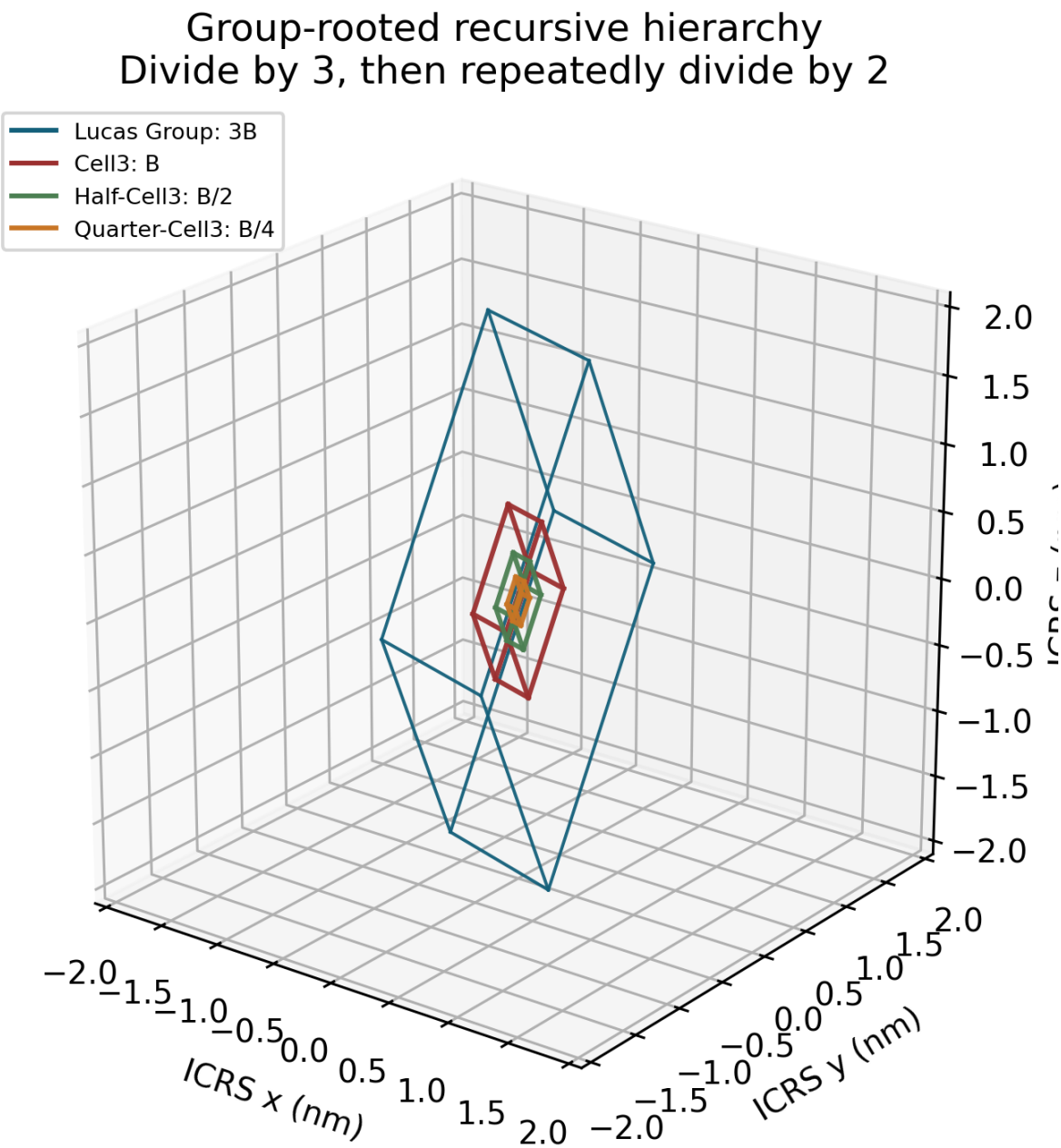


Figure 4. Nested group-rooted hierarchy. All nested cells share one center, one orientation, and one physical coordinate scale. The Lucas Group has basis $3B$; later cells have bases B , $B/2$, and $B/4$.

Depth n	Intervals per group axis	Subcells in group	Merged nodes
0	3	27	64

Depth n	Intervals per group axis	Subcells in group	Merged nodes
1	6	216	343
2	12	1,728	2,197
3	24	13,824	15,625
4	48	110,592	117,649

Every finite rung is discrete and exactly addressable. Because the dyadic depth can continue, the hierarchy has no terminal smallest rung. Shape remains fixed while scale changes.

5. The 27-node Lucas graph

Inside one centered Cell3, the first Lucas subdivision uses addresses (i,j,k) , where each coordinate is -1, 0, or 1. The physical point is $p = B(i,j,k)/2$. The 27 points consist of 8 corners, 12 edge midpoints, 6 face centers, and 1 body center.

Nearest neighbors differ by one address step in exactly one coordinate. Connecting those pairs produces 54 primitive graph links.

Complete 27-node Lucas graph 27 exact nodes and 54 primitive links

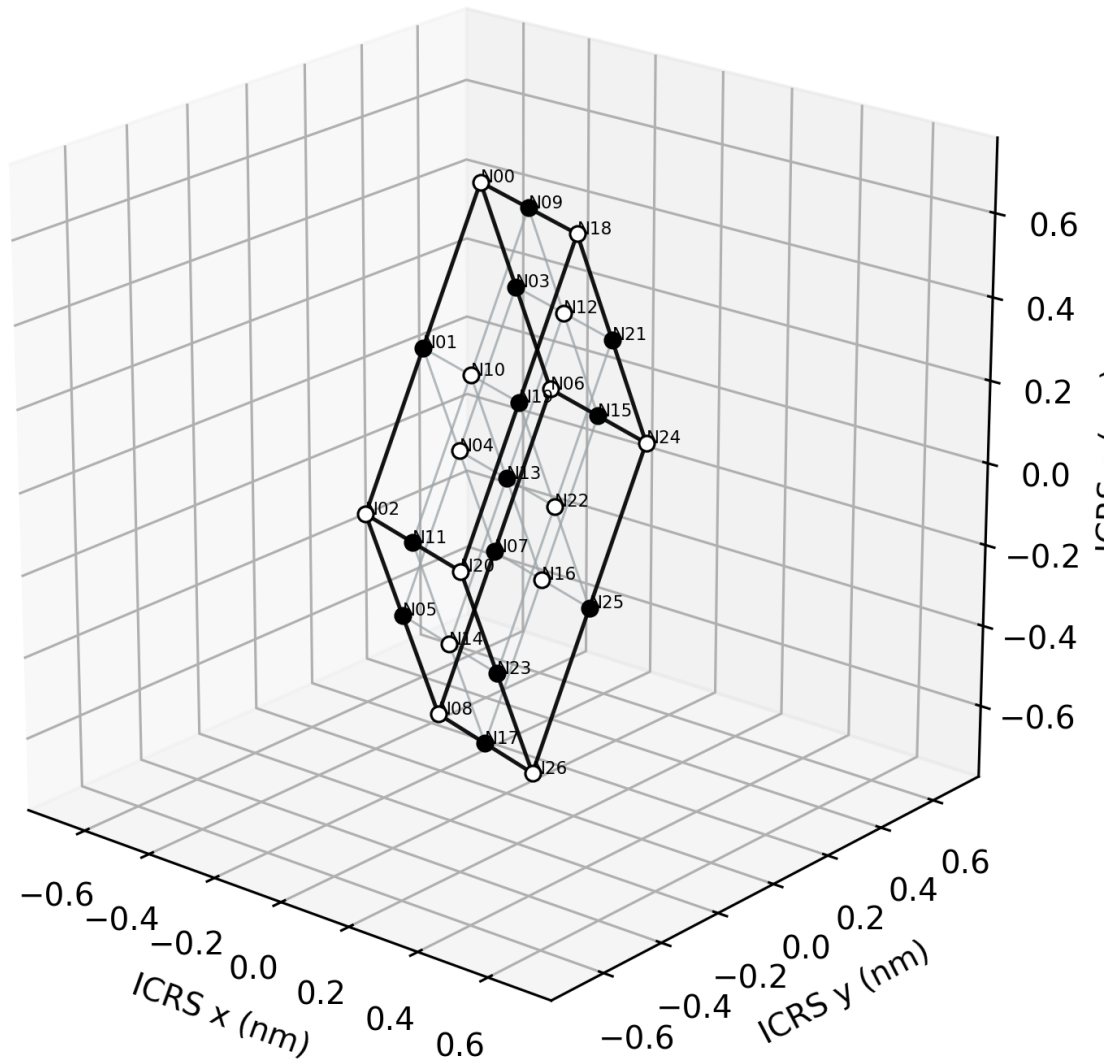


Figure 5. Complete 27-node Lucas graph with all 54 primitive links. The coordinate axes use equal nanometre scaling. Compact node IDs are used to prevent label overlap; Appendix B maps every ID to its full address and ICRS coordinates.

All 27 Lucas nodes in dimensionally controlled orthographic views
Filled = even lambda; open = odd lambda

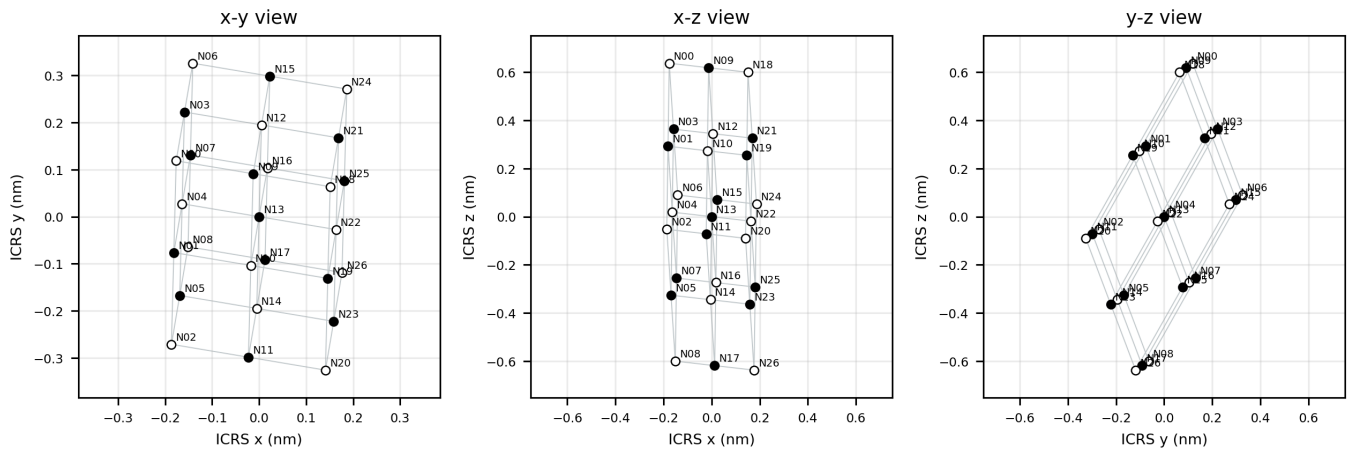


Figure 6. Three orthographic views of all 27 nodes. Each view uses equal physical scaling on its two axes. Compact node IDs correspond to the complete address ledger in Appendix B.

6. The seven sheets

The 27 nodes separate into seven parallel sheets according to the integer grade $\lambda = i + j + k$. The possible grades are -3, -2, -1, 0, 1, 2, and 3. Their populations are 1, 3, 6, 7, 6, 3, and 1.

The exact adjacent sheet spacing is $h = 0.151272688042$ nm. The positive sheet-normal direction is RA 3.082265 deg and Dec -28.568338 deg in ICRS. Adjacent sheets alternate parity according to $\chi = (-1)^\lambda$.

Lucas seven-sheet organization in an orthonormal physical frame
Normal direction: RA 3.082265 deg, Dec -28.568338 deg

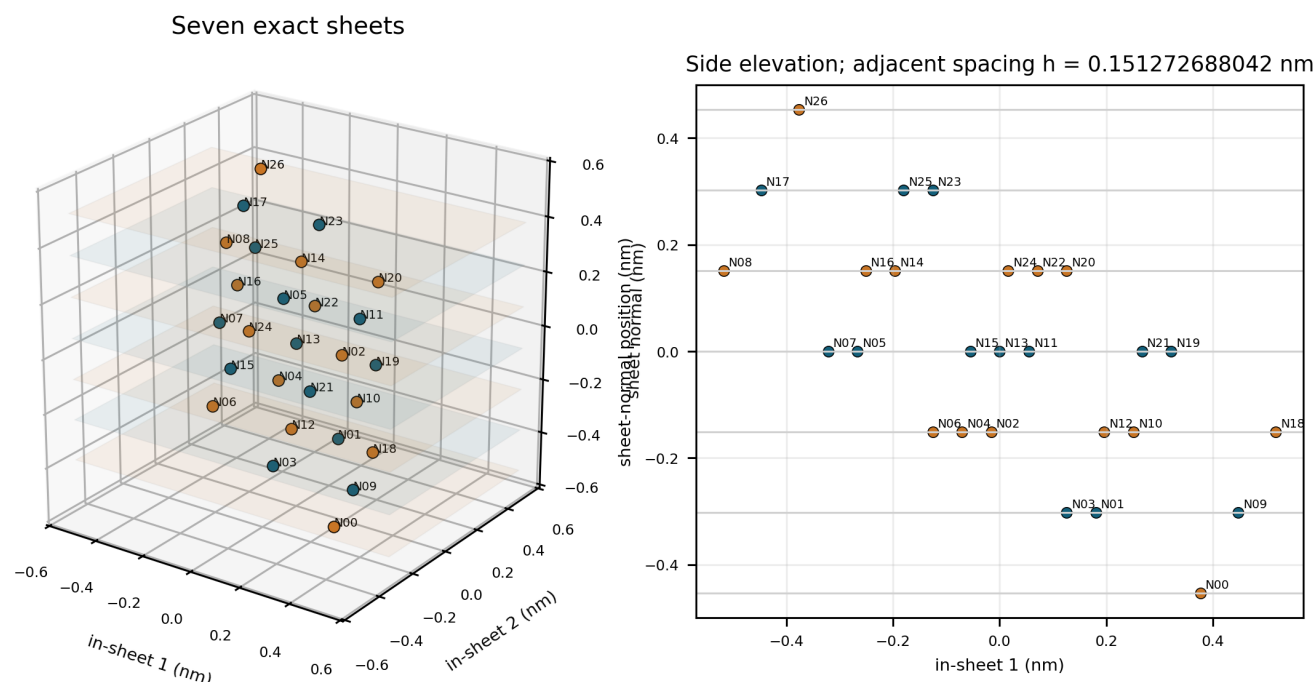


Figure 7. The seven-sheet organization shown in an orthonormal physical coordinate frame. The left panel displays the full sheet stack. The right panel is a side elevation that directly shows equal spacing. Both panels use nanometres and preserve physical scale.

λ	Population	Parity χ	Normal offset (nm)
-3	1	-1	-0.453818064127
-2	3	+1	-0.302545376085
-1	6	-1	-0.151272688042
0	7	+1	0.000000000000
1	6	-1	0.151272688042
2	3	+1	0.302545376085
3	1	-1	0.453818064127

The seven sheets are not extra surfaces added after the nodes are chosen. They are exact affine level sets of the address-sum rule. Their equal spacing follows from the inverse-transpose geometry of Cell3.

7. Graph paths and holonomy

A graph tells which nodes are connected. Geometry tells where the nodes are. The K-Field uses both. A path can move through several links and return to the starting node. If a phase, registry shift, or path-order memory remains after the loop closes, that remaining state is called holonomy.

The 27-node graph has incidence rank 26 and cycle-space dimension 28. This means that many independent loop histories can exist even though the graph has only 27 node positions. The cycle space stores relational history; the cut space describes node-visible changes such as occupation or divergence.

8. Cross-domain dimensional collapse

Cell3 has three positive face areas, but a common scale change multiplies all three by the same factor. Removing that common scale leaves two independent shape coordinates.

$$F = \ln(A_{ac}/A_{bc}) = -0.315073473002257$$

$$M = \ln(A_{bc}/A_{ab}) = 0.607672444153543$$

$$u = (F+M)/\sqrt{2} = 0.206898716669281; v = (M-F)/\sqrt{6} = 0.376709443211365$$

These two coordinates define a scale-free Cell3 shape point. Many scientific quantities can then be written as different functions of the same two coordinates. This is the meaning of cross-domain dimensional collapse: many observations can live in a large measurement space while still being generated by a small latent geometric object.

Channel	Cell3 formula	Computed value
fine channel alpha_N	$A_{ac}/(100 A_{bc})$	0.007297352564592
inverse fine channel	$100 A_{bc}/A_{ac}$	137.035999172106
mass channel eta_N	$1000 A_{bc}/A_{ab}$	1836.152673337283
reduced-mass factor rho	$\eta_N/(1+\eta_N)$	0.999455679424735

Different experimental routes can independently constrain the same hidden coordinate. Experimental independence is therefore not the same as latent dimensional independence. Several experiments can measure one coordinate without creating additional geometric dimensions.

9. The hidden one-plus-two kernel

The same mathematical split appears repeatedly: one common coordinate plus two relative coordinates. In face geometry, the one coordinate is common scale and the two coordinates are u and v. In Lucas addresses, the one coordinate is sheet grade lambda and the two coordinates describe same-sheet motion. In three-branch phase space, one common phase is physically redundant and two relative phases remain.

Representation	One common dimension	Two relative dimensions
face areas	common scale	shape coordinates u and v
Lucas addresses	sheet quotient lambda	two A2 same-sheet directions
branch phases	common phase gauge	two relative phase coordinates
weighted spectrum	coherent branch mode	soft and hard internal modes

The physical metric couples these algebraic parts. The documented quotient-fiber coupling is $\rho_{QF}^2 = 0.543701653538894$, with renormalizer $\chi_{QF} = 2.191548594808$. The kernel is therefore one-plus-two in structure but not physically separable into three unrelated pieces.

10. Mass, light, and gravity branches

Within the K-Field framework, matter is treated as stable occupation in the recursive graph. Mass measures resistance to changing that occupied state. Light-related and atomic channels are generated from the Cell3 face

ratios. Gravity uses occupancy, reciprocal propagation geometry, and a local directional response compiler.

The native gravity scalar is $\text{Gamma2} = 2.400587822460640964\text{e-}43$. The pre-directional rendered scalar is $G_{\text{base}} = 6.674239694206957015\text{e-}11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

The propagation and response metrics do not commute. Their generalized eigenproblem produces three relative response channels:

Mode	Generalized eigenvalue lambda	Relative amplitude 1/sqrt(lambda)	Interpretation
1	1.176761003328	0.921840673261	soft / enhanced
2	3.080568112610	0.569750338885	intermediate
3	18.035113207371	0.235472700023	hard / suppressed

These are not extra spatial dimensions. They are principal response directions created by comparing two noncommuting metric layers. The same underlying three-dimensional geometry can therefore respond differently along different geometric combinations.

11. Smallest complete educational summary

The smallest complete K-Field kernel contains: one authoritative Cell3; three reciprocal plane families; one 3 x 3 x 3 Lucas Group; one 27-node, 54-link local graph; seven equally spaced sheets; two scale-free shape coordinates; one common scale or quotient coordinate; and physical branch maps that render mass, fine-structure, phase, and gravity-related observables.

The most important lesson is that complicated behavior can arise from a small geometric grammar. Cell3 supplies the local shape. The Lucas Group supplies collective organization. The nodes and links supply discrete paths. The sheets supply parity and phase layers. The two hidden shape coordinates connect many domains to one common geometric state.

Appendix A. Complete numerical ledger

Object	Value
a	0.334179679073 nm
b	0.586410890415 nm
c	0.792319765038 nm
angle ab	83.933483012313 deg
angle ac	80.448130018338 deg
angle bc	50.363231813756 deg
A_ab	0.194869165470 nm ²
A_ac	0.261106235800 nm ²
A_bc	0.357809539129 nm ²
det(B)	-0.117914891917 nm ³
V	0.117914891917 nm ³
a+b+c	1.343653621680 nm
u-family spacing	0.329546529709 nm
v-family spacing	0.451597379724 nm
w-family spacing	0.605097741515 nm
sheet spacing h	0.151272688042 nm
sheet normal RA	3.082265053899 deg
sheet normal Dec	-28.568337610690 deg
F	-0.315073473002257
M	0.607672444153543
u	0.206898716669281
v	0.376709443211365
alpha_N	0.007297352564592
1/alpha_N	137.035999172106
eta_N	1836.152673337283
rho	0.999455679424735

Appendix B. Complete 27-node Lucas ledger

ID	Address	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	lambda	chi
N00	(-1,-1,-1)	-0.176351350144	0.118682429904	0.637311498326	-3	-1
N01	(-1,-1,0)	-0.181416008957	-0.076211421278	0.292444294144	-2	+1
N02	(-1,-1,1)	-0.186480667770	-0.271105272459	-0.052422910037	-1	-1
N03	(-1,0,-1)	-0.158685962221	0.222342355125	0.363610955630	-2	+1
N04	(-1,0,0)	-0.163750621035	0.027448503943	0.018743751448	-1	-1
N05	(-1,0,1)	-0.168815279848	-0.167445347239	-0.326123452733	0	+1
N06	(-1,1,-1)	-0.141020574299	0.326002280346	0.089910412934	-1	-1
N07	(-1,1,0)	-0.146085233113	0.131108429164	-0.254956791248	0	+1
N08	(-1,1,1)	-0.151149891926	-0.063785422018	-0.599823995429	1	-1
N09	(0,-1,-1)	-0.012600729109	0.091233925961	0.618567746877	-2	+1
N10	(0,-1,0)	-0.017665387922	-0.103659925221	0.273700542696	-1	-1
N11	(0,-1,1)	-0.022730046735	-0.298553776403	-0.071166661485	0	+1
N12	(0,0,-1)	0.005064658813	0.194893851182	0.344867204181	-1	-1

ID	Address	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	lambda	chi
N13	(0,0,0)	0.000000000000	0.000000000000	0.000000000000	0	+1
N14	(0,0,1)	-0.005064658813	-0.194893851182	-0.344867204181	1	-1
N15	(0,1,-1)	0.022730046735	0.298553776403	0.071166661485	0	+1
N16	(0,1,0)	0.017665387922	0.103659925221	-0.273700542696	1	-1
N17	(0,1,1)	0.012600729109	-0.091233925961	-0.618567746877	2	+1
N18	(1,-1,-1)	0.151149891926	0.063785422018	0.599823995429	-1	-1
N19	(1,-1,0)	0.146085233113	-0.131108429164	0.254956791248	0	+1
N20	(1,-1,1)	0.141020574299	-0.326002280346	-0.089910412934	1	-1
N21	(1,0,-1)	0.168815279848	0.167445347239	0.326123452733	0	+1
N22	(1,0,0)	0.163750621035	-0.027448503943	-0.018743751448	1	-1
N23	(1,0,1)	0.158685962221	-0.222342355125	-0.363610955630	2	+1
N24	(1,1,-1)	0.186480667770	0.271105272459	0.052422910037	1	-1
N25	(1,1,0)	0.181416008957	0.076211421278	-0.292444294144	2	+1
N26	(1,1,1)	0.176351350144	-0.118682429904	-0.637311498326	3	-1

Closing statement

The K-Field provides a unified geometric language for studying scale, shape, node organization, reciprocal planes, phase, graph loops, and cross-domain physical witnesses. The numerical Cell3 basis fixes the geometry. Every figure and table in this reference follows from that basis without rescaling one coordinate axis differently from another.